

Class 10 Math — Tough Practice Set

100 Questions | Level: Tough

StudyShelf — studysshelf practice worksheet. Answer key provided at the end.

- Q1.** The product of two consecutive positive integers is 420. Form a quadratic equation and find the two integers.
- Q2.** A point P is at a distance of 16 cm from the centre of a circle of radius 3 cm. Find the length of the tangent drawn from P to the circle.
- Q3.** Show that the points A(9,8), B(6,2) and C(0,-10) are collinear.
- Q4.** A box contains 19 bulbs out of which 2 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.
- Q5.** Show that the points A(8,8), B(10,0) and C(12,-8) are collinear.
- Q6.** Find the mean of the following grouped data using the direct method: 0-10: freq 4; 10-20: freq 9; 20-30: freq 6; 30-40: freq 2; 40-50: freq 15
- Q7.** Find the mean of the following grouped data using the direct method: 0-10: freq 6; 10-20: freq 13; 20-30: freq 2; 30-40: freq 5; 40-50: freq 5
- Q8.** A person saves ₹804 in the first month and increases the saving by ₹140 every month. Find the total amount saved in 13 months.
- Q9.** Show that the points A(1,3), B(8,10) and C(22,24) are collinear.
- Q10.** In triangle ABC, DE is drawn parallel to BC, meeting AB at D and AC at E. If AD = 4 cm, DB = 6 cm and AE = 5 cm, find EC using the Basic Proportionality Theorem.
- Q11.** A point P is at a distance of 24 cm from the centre of a circle of radius 9 cm. Find the length of the tangent drawn from P to the circle.
- Q12.** The product of two consecutive positive integers is 506. Form a quadratic equation and find the two integers.
- Q13.** Show that the points A(10,2), B(5,1) and C(-5,-1) are collinear.
- Q14.** A solid is in the shape of a cylinder of radius 5 cm and height 14 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = \frac{22}{7}$)
- Q15.** A solid is in the shape of a cylinder of radius 3 cm and height 8 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = \frac{22}{7}$)
- Q16.** A box contains 11 bulbs out of which 3 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.
- Q17.** Find the mean of the following grouped data using the direct method: 0-10: freq 2; 10-20: freq 7; 20-30: freq 11; 30-40: freq 12; 40-50: freq 5
- Q18.** A solid is in the shape of a cylinder of radius 8 cm and height 7 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = \frac{22}{7}$)

Q19. Find the mean of the following grouped data using the direct method: 0-10: freq 2; 10-20: freq 3; 20-30: freq 7; 30-40: freq 4; 40-50: freq 4

Q20. Show that the points A(3,10), B(10,0) and C(17,-10) are collinear.

Q21. Show that the points A(8,0), B(1,4) and C(-6,8) are collinear.

Q22. A solid is in the shape of a cylinder of radius 8 cm and height 12 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)

Q23. The product of two consecutive positive integers is 1332. Form a quadratic equation and find the two integers.

Q24. A solid is in the shape of a cylinder of radius 7 cm and height 8 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)

Q25. From the top of a 20 m high tower, the angle of depression of a car is 30° , and from the base of the tower the angle of depression to another point further away is not applicable; instead: the angle of elevation of the top of the tower from a point A is 60° , and from a point B, 23.1 m further away in line with A, the angle of elevation is 30° . Find the height of the tower.

Q26. A box contains 16 bulbs out of which 3 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.

Q27. A point P is at a distance of 18 cm from the centre of a circle of radius 7 cm. Find the length of the tangent drawn from P to the circle.

Q28. A person saves ₹1527 in the first month and increases the saving by ₹207 every month. Find the total amount saved in 15 months.

Q29. The product of two consecutive positive integers is 1260. Form a quadratic equation and find the two integers.

Q30. A box contains 16 bulbs out of which 2 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.

Q31. Show that the points A(6,6), B(1,3) and C(-9,-3) are collinear.

Q32. A box contains 18 bulbs out of which 2 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.

Q33. Show that the points A(10,2), B(2,5) and C(-14,11) are collinear.

Q34. A person saves ₹1834 in the first month and increases the saving by ₹277 every month. Find the total amount saved in 13 months.

Q35. A person saves ₹1450 in the first month and increases the saving by ₹168 every month. Find the total amount saved in 10 months.

Q36. Show that the points A(3,5), B(10,4) and C(17,3) are collinear.

Q37. A person saves ₹1024 in the first month and increases the saving by ₹200 every month. Find the total amount saved in 18 months.

Q38. The product of two consecutive positive integers is 210. Form a quadratic equation and find the two integers.

Q39. The product of two consecutive positive integers is 110. Form a quadratic equation and find the two integers.

Q40. Show that the points A(10,4), B(9,0) and C(7,-8) are collinear.

Q41. Find the mean of the following grouped data using the direct method: 0-10: freq 14; 10-20: freq 5; 20-30: freq 15; 30-40: freq 15; 40-50: freq 2

Q42. Show that the points A(2,4), B(8,2) and C(20,-2) are collinear.

Q43. A person saves ₹750 in the first month and increases the saving by ₹141 every month. Find the total amount saved in 20 months.

Q44. A box contains 20 bulbs out of which 4 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.

Q45. A person saves ₹1989 in the first month and increases the saving by ₹144 every month. Find the total amount saved in 16 months.

Q46. A person saves ₹1624 in the first month and increases the saving by ₹147 every month. Find the total amount saved in 19 months.

Q47. Find the mean of the following grouped data using the direct method: 0-10: freq 3; 10-20: freq 2; 20-30: freq 2; 30-40: freq 5; 40-50: freq 9

Q48. A person saves ₹1944 in the first month and increases the saving by ₹196 every month. Find the total amount saved in 18 months.

Q49. A point P is at a distance of 11 cm from the centre of a circle of radius 6 cm. Find the length of the tangent drawn from P to the circle.

Q50. A box contains 14 bulbs out of which 4 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.

Q51. A solid is in the shape of a cylinder of radius 6 cm and height 13 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)

Q52. A box contains 18 bulbs out of which 3 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.

Q53. A solid is in the shape of a cylinder of radius 8 cm and height 14 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)

Q54. A point P is at a distance of 18 cm from the centre of a circle of radius 3 cm. Find the length of the tangent drawn from P to the circle.

Q55. From the top of a 50 m high tower, the angle of depression of a car is 30° , and from the base of the tower the angle of depression to another point further away is not applicable; instead: the angle of elevation of the top of the tower from a point A is 60° , and from a point B, 57.7 m further away in line with A, the angle of elevation is 30° . Find the height of the tower.

- Q56.** A solid is in the shape of a cylinder of radius 3 cm and height 7 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)
- Q57.** A box contains 15 bulbs out of which 5 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.
- Q58.** A person saves ■511 in the first month and increases the saving by ■119 every month. Find the total amount saved in 11 months.
- Q59.** Show that the points A(7,2), B(1,6) and C(-11,14) are collinear.
- Q60.** A person saves ■561 in the first month and increases the saving by ■231 every month. Find the total amount saved in 20 months.
- Q61.** Show that the points A(3,7), B(10,7) and C(24,7) are collinear.
- Q62.** A person saves ■790 in the first month and increases the saving by ■129 every month. Find the total amount saved in 19 months.
- Q63.** A solid is in the shape of a cylinder of radius 8 cm and height 15 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)
- Q64.** A point P is at a distance of 9 cm from the centre of a circle of radius 4 cm. Find the length of the tangent drawn from P to the circle.
- Q65.** From the top of a 40 m high tower, the angle of depression of a car is 30° , and from the base of the tower the angle of depression to another point further away is not applicable; instead: the angle of elevation of the top of the tower from a point A is 60° , and from a point B, 46.2 m further away in line with A, the angle of elevation is 30° . Find the height of the tower.
- Q66.** Show that the points A(9,9), B(9,4) and C(9,-6) are collinear.
- Q67.** Find the mean of the following grouped data using the direct method: 0-10: freq 14; 10-20: freq 14; 20-30: freq 4; 30-40: freq 5; 40-50: freq 14
- Q68.** A person saves ■1489 in the first month and increases the saving by ■197 every month. Find the total amount saved in 20 months.
- Q69.** A solid is in the shape of a cylinder of radius 6 cm and height 12 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)
- Q70.** A person saves ■1756 in the first month and increases the saving by ■191 every month. Find the total amount saved in 13 months.
- Q71.** A person saves ■1977 in the first month and increases the saving by ■174 every month. Find the total amount saved in 15 months.
- Q72.** A box contains 12 bulbs out of which 3 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.
- Q73.** A person saves ■1881 in the first month and increases the saving by ■190 every month. Find the total amount saved in 16 months.
- Q74.** Show that the points A(0,5), B(4,9) and C(12,17) are collinear.

- Q75.** A solid is in the shape of a cylinder of radius 7 cm and height 12 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)
- Q76.** A solid is in the shape of a cylinder of radius 7 cm and height 9 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)
- Q77.** The product of two consecutive positive integers is 702. Form a quadratic equation and find the two integers.
- Q78.** A box contains 11 bulbs out of which 5 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.
- Q79.** A solid is in the shape of a cylinder of radius 8 cm and height 10 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)
- Q80.** A solid is in the shape of a cylinder of radius 7 cm and height 14 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)
- Q81.** The product of two consecutive positive integers is 380. Form a quadratic equation and find the two integers.
- Q82.** A person saves ₹918 in the first month and increases the saving by ₹292 every month. Find the total amount saved in 14 months.
- Q83.** Find the mean of the following grouped data using the direct method: 0-10: freq 3; 10-20: freq 4; 20-30: freq 11; 30-40: freq 9; 40-50: freq 10
- Q84.** A solid is in the shape of a cylinder of radius 7 cm and height 13 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)
- Q85.** Show that the points A(10,3), B(0,5) and C(-20,9) are collinear.
- Q86.** The product of two consecutive positive integers is 1190. Form a quadratic equation and find the two integers.
- Q87.** Show that the points A(4,2), B(0,8) and C(-4,14) are collinear.
- Q88.** Find the mean of the following grouped data using the direct method: 0-10: freq 12; 10-20: freq 2; 20-30: freq 7; 30-40: freq 8; 40-50: freq 9
- Q89.** The product of two consecutive positive integers is 30. Form a quadratic equation and find the two integers.
- Q90.** Show that the points A(4,1), B(1,0) and C(-2,-1) are collinear.
- Q91.** Find the mean of the following grouped data using the direct method: 0-10: freq 10; 10-20: freq 9; 20-30: freq 15; 30-40: freq 4; 40-50: freq 8
- Q92.** A point P is at a distance of 12 cm from the centre of a circle of radius 5 cm. Find the length of the tangent drawn from P to the circle.
- Q93.** A person saves ₹938 in the first month and increases the saving by ₹190 every month. Find the total amount saved in 19 months.

Q94. Find the mean of the following grouped data using the direct method: 0-10: freq 10; 10-20: freq 15; 20-30: freq 12; 30-40: freq 11; 40-50: freq 5

Q95. A person saves ₹1207 in the first month and increases the saving by ₹104 every month. Find the total amount saved in 16 months.

Q96. A box contains 12 bulbs out of which 2 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.

Q97. Show that the points A(5,6), B(8,4) and C(14,0) are collinear.

Q98. The product of two consecutive positive integers is 756. Form a quadratic equation and find the two integers.

Q99. A box contains 10 bulbs out of which 2 are defective. One bulb is drawn at random. Find the probability that it is (i) defective (ii) not defective.

Q100. A solid is in the shape of a cylinder of radius 7 cm and height 10 cm, surmounted by a hemisphere of the same radius. Find the total volume of the solid. (use $\pi = 22/7$)

Answer Key

1. $x = 20$ and $x+1 = 21$
2. Tangent length ≈ 15.72 cm
3. Area of triangle ABC = 0, hence the points are collinear.
4. $P(\text{defective}) = 2/19$, $P(\text{not defective}) = 17/19$
5. Area of triangle ABC = 0, hence the points are collinear.
6. Mean ≈ 29.17
7. Mean ≈ 21.77
8. Total savings = ■21372
9. Area of triangle ABC = 0, hence the points are collinear.
10. EC = 7.5 cm
11. Tangent length ≈ 22.25 cm
12. $x = 22$ and $x+1 = 23$
13. Area of triangle ABC = 0, hence the points are collinear.
14. Total volume $\approx 1361.9 \text{ cm}^3$
15. Total volume $\approx 282.86 \text{ cm}^3$
16. $P(\text{defective}) = 3/11$, $P(\text{not defective}) = 8/11$
17. Mean ≈ 27.97
18. Total volume $\approx 2480.76 \text{ cm}^3$
19. Mean ≈ 27.50
20. Area of triangle ABC = 0, hence the points are collinear.
21. Area of triangle ABC = 0, hence the points are collinear.
22. Total volume $\approx 3486.48 \text{ cm}^3$
23. $x = 36$ and $x+1 = 37$
24. Total volume $\approx 1950.67 \text{ cm}^3$
25. Height ≈ 20 m (verify: distance AB ≈ 23.09 m)
26. $P(\text{defective}) = 3/16$, $P(\text{not defective}) = 13/16$
27. Tangent length ≈ 16.58 cm
28. Total savings = ■44640
29. $x = 35$ and $x+1 = 36$
30. $P(\text{defective}) = 2/16$, $P(\text{not defective}) = 14/16$
31. Area of triangle ABC = 0, hence the points are collinear.
32. $P(\text{defective}) = 2/18$, $P(\text{not defective}) = 16/18$
33. Area of triangle ABC = 0, hence the points are collinear.
34. Total savings = ■45448
35. Total savings = ■22060
36. Area of triangle ABC = 0, hence the points are collinear.
37. Total savings = ■49032

38. $x = 14$ and $x+1 = 15$
39. $x = 10$ and $x+1 = 11$
40. Area of triangle ABC = 0, hence the points are collinear.
41. Mean ≈ 22.25
42. Area of triangle ABC = 0, hence the points are collinear.
43. Total savings = ■41790
44. $P(\text{defective}) = 4/20$, $P(\text{not defective}) = 16/20$
45. Total savings = ■49104
46. Total savings = ■55993
47. Mean ≈ 32.14
48. Total savings = ■64980
49. Tangent length ≈ 9.22 cm
50. $P(\text{defective}) = 4/14$, $P(\text{not defective}) = 10/14$
51. Total volume $\approx 1923.43 \text{ cm}^3$
52. $P(\text{defective}) = 3/18$, $P(\text{not defective}) = 15/18$
53. Total volume $\approx 3888.76 \text{ cm}^3$
54. Tangent length ≈ 17.75 cm
55. Height ≈ 50 m (verify: distance AB ≈ 57.74 m)
56. Total volume $\approx 254.57 \text{ cm}^3$
57. $P(\text{defective}) = 5/15$, $P(\text{not defective}) = 10/15$
58. Total savings = ■12166
59. Area of triangle ABC = 0, hence the points are collinear.
60. Total savings = ■55110
61. Area of triangle ABC = 0, hence the points are collinear.
62. Total savings = ■37069
63. Total volume $\approx 4089.9 \text{ cm}^3$
64. Tangent length ≈ 8.06 cm
65. Height ≈ 40 m (verify: distance AB ≈ 46.19 m)
66. Area of triangle ABC = 0, hence the points are collinear.
67. Mean ≈ 23.24
68. Total savings = ■67210
69. Total volume $\approx 1810.29 \text{ cm}^3$
70. Total savings = ■37726
71. Total savings = ■47925
72. $P(\text{defective}) = 3/12$, $P(\text{not defective}) = 9/12$
73. Total savings = ■52896
74. Area of triangle ABC = 0, hence the points are collinear.
75. Total volume $\approx 2566.67 \text{ cm}^3$
76. Total volume $\approx 2104.67 \text{ cm}^3$

77. $x = 26$ and $x+1 = 27$
78. $P(\text{defective}) = 5/11$, $P(\text{not defective}) = 6/11$
79. Total volume $\approx 3084.19 \text{ cm}^3$
80. Total volume $\approx 2874.67 \text{ cm}^3$
81. $x = 19$ and $x+1 = 20$
82. Total savings = ■39424
83. Mean ≈ 30.14
84. Total volume $\approx 2720.67 \text{ cm}^3$
85. Area of triangle ABC = 0, hence the points are collinear.
86. $x = 34$ and $x+1 = 35$
87. Area of triangle ABC = 0, hence the points are collinear.
88. Mean ≈ 25.00
89. $x = 5$ and $x+1 = 6$
90. Area of triangle ABC = 0, hence the points are collinear.
91. Mean ≈ 23.04
92. Tangent length $\approx 10.91 \text{ cm}$
93. Total savings = ■50312
94. Mean ≈ 22.36
95. Total savings = ■31792
96. $P(\text{defective}) = 2/12$, $P(\text{not defective}) = 10/12$
97. Area of triangle ABC = 0, hence the points are collinear.
98. $x = 27$ and $x+1 = 28$
99. $P(\text{defective}) = 2/10$, $P(\text{not defective}) = 8/10$
100. Total volume $\approx 2258.67 \text{ cm}^3$